ALGEBRA 1 · UNIT 6: SYSTEMS OF EQUATIONS AND INEQUALITIES

Drone Delivery Zones

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students work for a drone delivery company in the town of Riverton. They test which houses fall inside zones defined by inequalities, write inequalities from graphs, and design new zones (including a diamond-shaped battery range for Honors) so every house is served.

This project uses points (houses) to give immediate meaning to test points and boundary lines. Students must decide whether a house on a boundary is served (solid vs. dashed), and then reverse the process to write inequalities that include or exclude chosen points.

Standards Addressed

- A-REI.12 Graph the solutions to a linear inequality in two variables as a half-plane
- A-CED.3 Represent constraints by inequalities
- A-CED.2 Create equations in two variables
- Honors: systems of inequalities and absolute value regions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: Which houses do you think are in the zone? Part 1: test points and graph Zone 1.
Day 2	Part 2: more zones, boundary cases, and finding gaps.
Day 3	Part 3: design new zones; create the final map.

Materials

- Student handout with the town map
- Colored pencils and rulers
- Blank coordinate grids
- Optional: Desmos to check zones

How the Three Versions Differ

Level	What changes
CP2	Given inequalities; students test houses, graph the zones, and design one new zone.
CP1	Students write inequalities from a graph, find houses in two zones at once, and design zones through chosen points.
Honors	Zones are systems (a triangle and a taxicab-distance diamond $\|x\| + \|y\| \le 8$); students design a region that includes some houses and excludes others.

Watch For (Common Misconceptions)

- Thinking a point on a dashed line is included.
- Shading above for every '>' even when the inequality is not solved for y (e.g., $-y > \ldots$).
- Graphing $x \le -3$ as a horizontal line.

Extension Ideas

- Add a no-fly zone over a stadium and route around it.
- Use real map coordinates of the school neighborhood.
- Count how many houses are in each zone and discuss fair workloads.

Answer Key – CP2

Part 1: Zone 1

#	Task	Answer
1	Is house A (2, 3) in Zone 1? Substitute to check.	$3 \geq 0$ is true, so yes.
2	Test every house. Write yes or no.	Yes: A, B, D, E, F, J. No: C, G, H, K.
3	Graph Zone 1 on the town map. Use a solid or dashed line and shade.	Solid line $y = x - 2$, shaded above (the side with (0, 0)).
4	Do your graph and table agree? Explain.	Yes; the houses above the line are the ones that tested true.

Part 2: More Zones

#	Task	Answer
5	Is the boundary of Zone 2 solid or dashed? House C (6, −2) is exactly on this line. Does it get a delivery from Drone Blue?	Dashed (<). For C: $-\frac{1}{2}(6) + 1 = -2$, and $-2 < -2$ is false, so C is not served by Drone Blue.
6	List the houses in Zone 2. Show your checks.	D ($-4 < 2.5$), F ($-1 < 4$), G ($-5 < 0.5$), J ($1 < 2$). Not A, B, C, E, H, K.
7	Graph Zone 3 on the grid. Which houses are in Zone 3?	Solid vertical line $x = -3$, shade left. Houses B, D, F.
8	Which houses are not in any zone?	C, H, and K

Part 3: Fill the Gaps

#	Task	Answer
9	Write one inequality for Zone 4. Show that each unserved house makes it true.	Example: $x \geq 4$. C (6), H (8), K (4) all work.
10	Graph Zone 4 and explain why you chose a solid or dashed line.	Example: solid so that K (4, 0) on the line is included.
11	Make a final list showing which zone(s) serve each house.	Answers vary.

Answer Key – CP1

Part 1: Read the Zones

#	Task	Answer
1	Write the inequality for Zone P (blue). Explain how you chose the symbol.	Dashed line through (0, 4) and (4, 0) with shading below: $y < -x + 4$
2	Write the inequality for Zone Q (orange).	Solid line through (0, −6) and (3, 0), shading above: $y \geq 2x - 6$
3	Which houses are in BOTH zones? Show your checks.	B (5 < 8; 5 ≥ −14), D (−4 < 7; −4 ≥ −12), F (−1 < 10; −1 ≥ −18), J (1 < 6; 1 ≥ −10)
4	Houses C and K lie exactly on the boundary of Zone P. Are they served by Zone P? Explain.	No. C: −2 < −2 is false. K: 0 < 0 is false. Points on a dashed line are not included.

Part 2: No-Fly Rules

#	Task	Answer
5	Drones may not fly over the stadium, which is everywhere with $y > 7$. Write the inequality for where drones may fly and graph it.	$y \leq 7$, solid horizontal line, shade below.
6	A drone flies along the river path from (−6, −9) to (3, 9). Write the equation of the river line, then write the inequality for the region to the…	Slope = 18 ÷ 9 = 2; $y = 2x + 3$. Region below the river: $y < 2x + 3$.
7	Which houses are on the right side of the river? Show your checks.	Check $y < 2x + 3$: A (3 < 7) yes, C yes, E (6 < 13) yes, G yes, H yes, K yes; B (5 < −5) no, D (−4 < −3) yes, F (−1 < −9) no, J (1 < −1) no. Right side: A, C, D, E, G, H, K.
8	Write an inequality that describes houses with x-coordinates between −5 and 5. Which houses are in that strip?	$-5 \leq x \leq 5$: A, B, D, E, G, J, K (C, F, and H are outside).

Part 3: Design Full Coverage

#	Task	Answer
9	Design two new zones. Each boundary must pass through two houses or the hub (0, 0). Write each inequality.	Answers vary.
10	Graph your zones and make a table showing each house's zone(s).	Answers vary.
11	Explain one choice of solid vs. dashed line in your design.	Answers vary.

Answer Key – Honors

Part 1: The Express Zone

#	Task	Answer
1	Write the equation of each side of the triangle.	$y = -2$; through $(-2, -2)$ and $(2, 6)$: $y = 2x + 2$; through $(6, -2)$ and $(2, 6)$: $y = -2x + 10$
2	Write a system of three inequalities for the Express Zone. Use a test point to choose each symbol.	$y \geq -2$, $y \leq 2x + 2$, $y \leq -2x + 10$ (test point $(2, 0)$ is inside)
3	Which houses are in the Express Zone? Show your checks.	A $(2, 3)$: $3 \geq -2$, $3 \leq 6$, $3 \leq 6 \rightarrow$ in. C $(6, -2)$: a vertex $\rightarrow$ in. K $(4, 0)$: $0 \leq 10$, $0 \leq 2 \rightarrow$ in. Others fail. Houses: A, C, K.
4	Write a system for the same triangle with the boundaries NOT included. Which house changes?	$y > -2$, $y < 2x + 2$, $y < -2x + 10$. House C (a vertex) is no longer included.

Part 2: Battery Range

#	Task	Answer
5	Write the battery rule as an inequality. Test houses A, B, and C.	$\lvert x\rvert + \lvert y\rvert \leq 8$. A: 5 yes. B: 9 no. C: 8 yes.
6	In Quadrant I, $\lvert x\rvert + \lvert y\rvert \leq 8$ becomes $x + y \leq 8$. Write the inequality for each of the other three quadrants.	QII ($x \leq 0$, $y \geq 0$): $-x + y \leq 8$. QIII: $-x - y \leq 8$. QIV: $x - y \leq 8$.
7	Graph the region $\lvert x\rvert + \lvert y\rvert \leq 8$. What shape is it? List every house inside it.	A diamond (square turned 45°) with vertices $(8, 0)$, $(0, 8)$, $(-8, 0)$, $(0, -8)$. Inside: A, C, D, F, G, J, K. Outside: B, E, H.
8	Which houses are inside the battery range but outside the Express Zone?	D, F, G, J

Part 3: Design a Second Hub

#	Task	Answer
9	Choose a location (h, k) for the new hub. Write its battery range as $\lvert x - h\rvert + \lvert y - k\rvert \leq 8$.	Answers vary. Example: $(5, 6)$.
10	Show that every house is within range of at least one hub.	Example $(5, 6)$: B: $\lvert -4 - 5\rvert + \lvert 5 - 6\rvert = 10$ (too far!). So try $(2, 5)$: B: $6 + 0 = 6$, E: $3 + 1 = 4$, H: $6 + 2 = 8$. All of B, E, H are covered.
11	Graph both diamonds and explain how you chose (h, k).	Answers vary.

Scoring note: In Part 3 the first example is intentionally shown failing to model the revision process; any hub that covers B, E, and H works (for example $(2, 5)$).

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	New zones cover every unserved house, with correct boundaries and shading. Goes beyond the minimum with a creative, realistic choice.	New zones cover every unserved house, with correct boundaries and shading.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Zone map shows every zone as a graphed inequality, lists the houses each zone serves, and shows full coverage. Polished, organized, and visually engaging.	Zone map shows every zone as a graphed inequality, lists the houses each zone serves, and shows full coverage. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.